



Department of Information Technology
Academic Year 2025 - 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CS3591 & Computer Networks

Name of the Faculty member (s): Mrs.M. Thulasi devi AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Domain Name Space (DNS)

- **Course Outcome:** CO1

- **Topic Learning Outcome:** TLO10

- **Activity Chosen:** Mind map

- **Justification:**

Within the first unit of the curriculum, the Domain Name System represents a crucial concept frequently featured in university examinations. Mind mapping serves as a hierarchical visualization tool that effectively demonstrates interconnections between various components of a comprehensive topic. This technique typically centers around a core concept, positioned centrally on a blank canvas, with related concepts, terminology, and visual representations radiating outward. This approach enables learners to grasp concepts more intuitively through visual organization and structured information presentation.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- Initial instruction session covering DNS fundamentals
- Fifteen-minute time allocation for the exercise
- Assignment given to learners to explore DNS concepts and classifications independently
- Follow-up session where participants were asked to demonstrate retention by creating visual mind maps
- Successful completion of the activity by students, with representative samples illustrated in Figures 1, 2, and 3

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	3	2	2	3	1	2	2	1	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	2	2	3	1	2	2	1	2	1	1	3
Justification for correlation	The students could be able to use and apply basic mathematics knowledge and fundamental programming knowledge in the different types of networks	The students could be able to identify basic engineering problem like shortest path finding in network and provide solution using mathematical and engineering science with various layer functions	The students could be able to design the various design strategies for engineering problems	The students could be able to interpret the application of various routing algorithms	The students could be able to use various frameworks for application layer	The students will complete the assessments in an ethical way	The students will work as a team and as an individual and demonstrate communication, problem-solving skills in networks	The students will work as a team and as an individual and demonstrate communication, problem-solving skills in networks	The students will be able to identify the task required to build various networks	The student will identify the technological advances made in the past and update the layers in OSI Model	The students could be able to apply computing skills to different application layer protocols.	The students could be able to provide the IT solution for various layer functions in the networking	The students could be able to develop software application for application layer.

- Images / Screenshot of the practice:

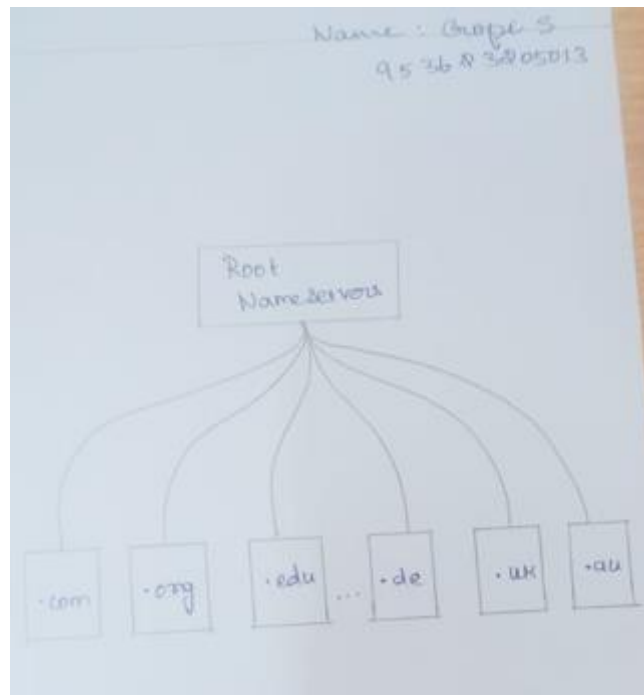


Figure 1: DNS Mindmap of S.Gopi

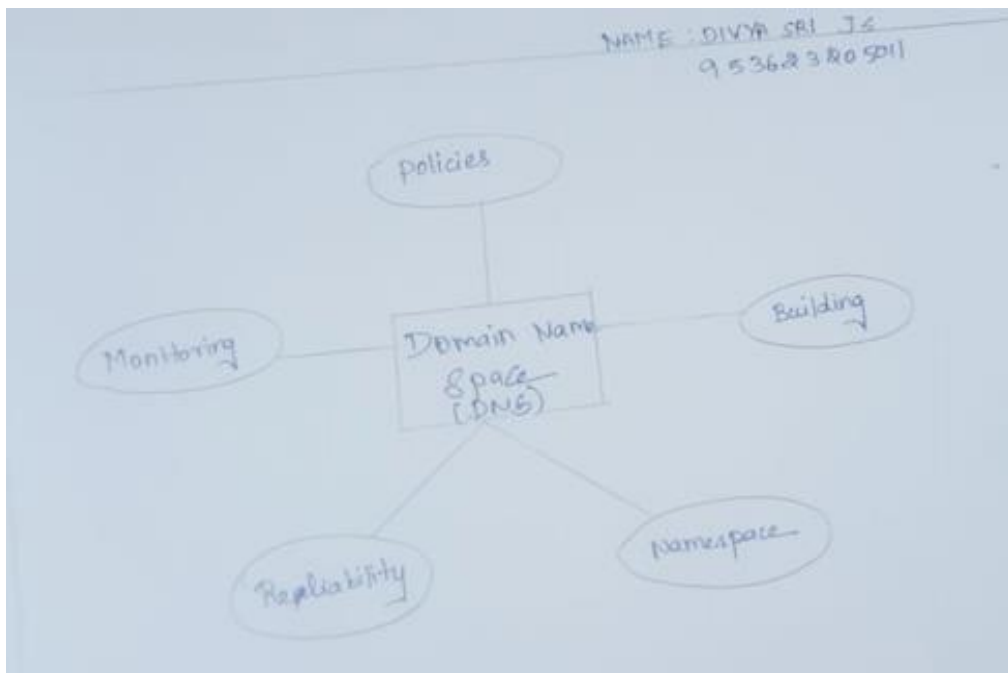


Figure 2: DNS Mind map of Divya Sri J S

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Participants indicated enhanced ability to remember and recall DNS-related concepts
- Learners noted that the exercise facilitated comprehensive review of previously covered material and strengthened autonomous learning capabilities
- The activity successfully conveyed DNS principles to students

- ❖ ***Benefit of the practice:***

- Strengthens analytical and problem-solving competencies while improving retention of DNS concepts
- Provides learners with opportunities to express individual perspectives and understanding through personalized mind map creation

- ❖ ***Challenges faced in implementation:***

- Several participants experienced difficulty in structuring their mind maps and determining appropriate organizational hierarchies
- Clarity issues were observed among slower-paced learners, which were addressed through individualized explanations, resulting in improved comprehension

References:

- ❖ <https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/4.Mindmap%20Unit%204.pdf>
- ❖ <https://equiip.eu/activity/activity-5-mind-mapping/>

Signature of Faculty Member

HOD